Paper 14.25 - Toolkit for Boundary-Repair Curvature

CQE-paper-14.25
CQECMPLX Formal Paper Series
Review-facing PDF built 2026-06-12

Construction Basis

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-14.25.md

Paper 14.25 - Toolkit for Boundary-Repair Curvature

Purpose

This support paper exposes the tools for Paper 14. The proof is the repair ledger in Paper 14. The toolkit shows how to reproduce and inspect it.

Digital Route

Run:

```
python production/formal-papers/CQE-paper-14/verify_boundary_repair_curvature.py
```

The verifier checks:

```
typed transport obligations
proof-boundary presence
zero repair for demonstrated rows
positive repair for open lifts
exact Paper 13 zero-repair reference
Cayley-Dickson/Oloid 1,8,8,1 normal form
dual-path oloid three-dyad coherence
```

Analog Route

Draw a route as a line. If it closes with no adjustment, mark repair score `0`. If it requires a named adjustment, write the adjustment as an obligation card.

Use the scoring ladder:

```
demonstrated = 0
bounded_local = 1
bounded_external = 2
registered_landing_forms = 3
open = 4
```

The analog curved carrier is a folded or rolling strip with three possible dyads: podal, antipodal, and shared.

Hidden-Guess Diagnostic

Before revealing the receipt, ask:

```
Is this row demonstrated, bounded, registered, or open?
Does it score zero repair or positive repair?
Is the claim substrate-level or physical-GR-level?
```

Then reveal the ledger row. The important training signal is whether the evaluator distinguishes closed repair accounting from physical curvature.

Boundary

This toolkit proves no Riemann, Ricci, Einstein, or measured gravitational claim. It exposes the repair ledger that a later physics interpretation must calibrate.